

QP Code : **NP-18696**

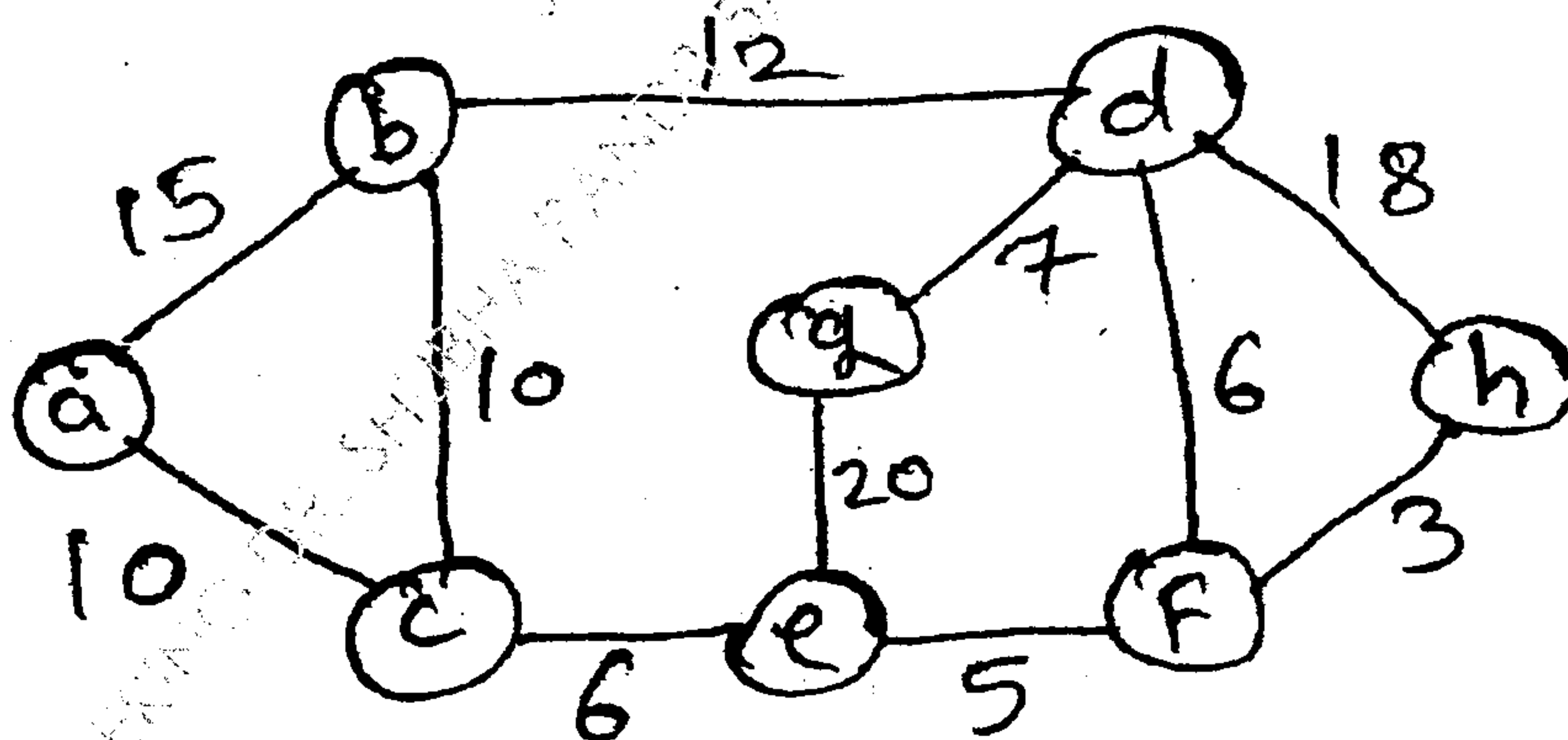
(3 Hours)

[Total Marks : 80

- N. B. :** (1) Question No. 1 is **compulsory**.
 (2) Attempt any **three** from **remaining** questions.
 (3) **Assume** suitable **data** if **necessary**.

1. (a) What is stack ? Give applications of it. 2
- (b) What is time complexity ? Determine time complexity of following code :- 3

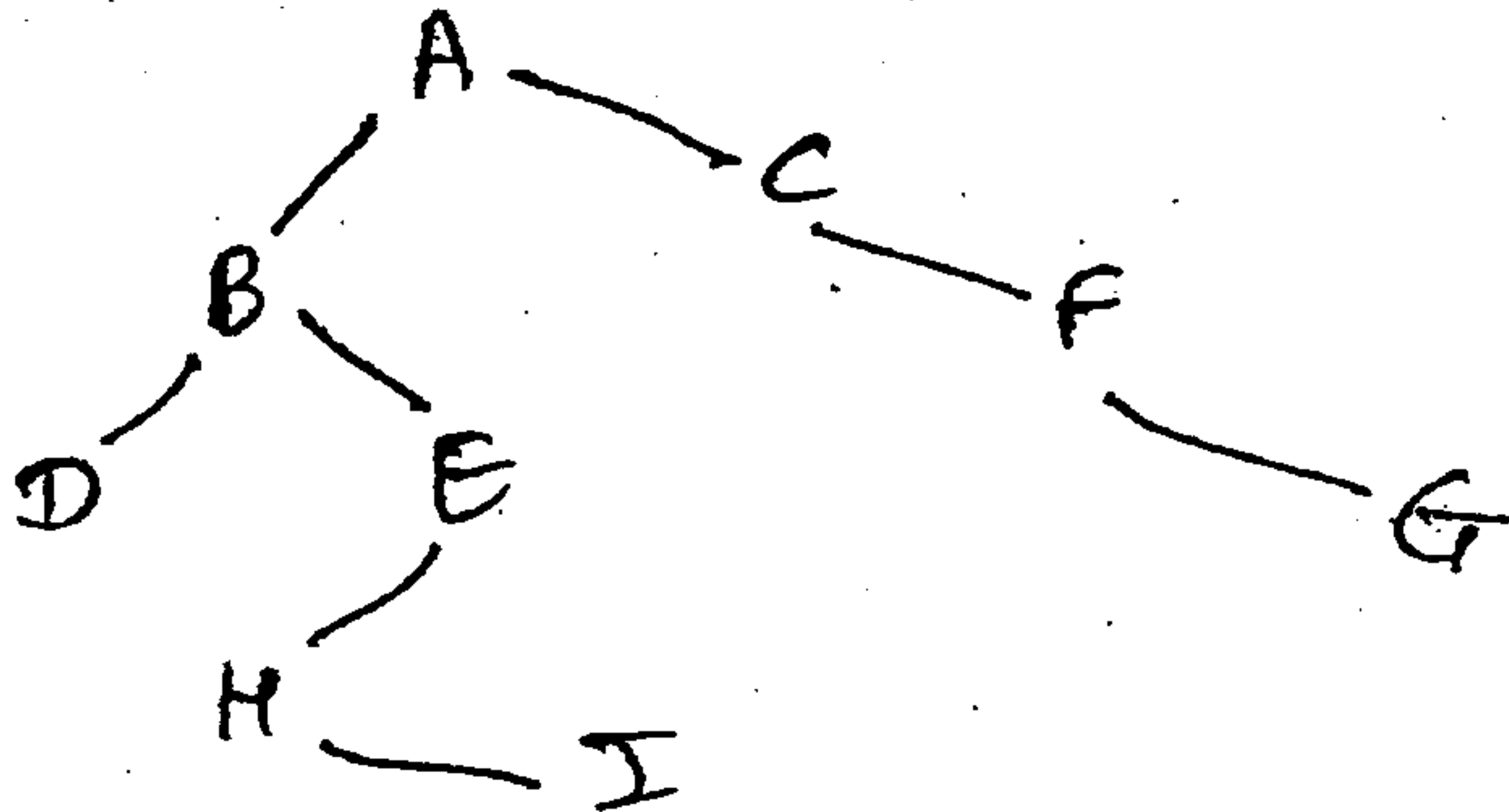
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for (i=1; i<=n; i++)
  for (j=1; j <=n; j++)
    x=x+1;
```
- (c) Explain with e.g. :- 3
 - (i) Complete binary tree
 - (ii) Degree of tree
 - (iii) Height of tree
- (d) Explain linked list with its various types. 3
- (e) Define double ended queue and give its applications. 3
- (f) Define asymptotic notation along with example. 3
- (g) Define Graph. List its types with example. 3
2. (a) Find the shortest path using Dijkstra's algorithm :- 10



- (b) Implement quick sort with example and find its complexity. 10
3. (a) Explain BFS and DFS with algorithm and proper example. 10
- (b) What is linked list ? Write 'C' function for insertion of 'n' elements. 10

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4. (a) Traverse the following binary tree into preorder, inorder, postorder by giving its algorithm. 10



- (b) Using Prim's algorithm find minimum spanning tree of a graph with example. Write algorithm of it. 10

5. (a) What is priority queue ? Give implementation of it. 10
 (b) What is graph ? Give representation of graph with example. State applications of it. 10

6. Solve any four :- 20

- (a) AVL Tree
 (b) Euclids algorithm
 (c) Sparse matrix
 (d) B-Tree
 (e) Circular linked list